
[Z8]
[Z7]
[Z6]
[Z5]
[Z4]
[Z3]
[Z2]
[Z1]
SEC-01 // ENCODER ARRAY
AN/SMC-925
TRACK
HOLD STEP
KNOB BANK
K7
LEN 1–64
K8
⇧+K8 SPEED
T: STEPS [1-64]
S: TRIG COND
B: MIDI CH [1-16]
T: RATE [⇧SPEED]
S: STEP TIE
B: SYNC [±120ms]
SPI2 · 595 MUX
K5
PROB %
K6
RATCH DIV
T: PROB [0-100%]
S: STEP PROB
B: — [UNASSIGNED]
T: RATCHET RATE
S: STEP RATCHET
B: TRACK COLOUR
QUAD DEC · ISR
K3
NR:BPM ⇧RT
K4
NR:LTC ⇧SC
T: BASE NOTE
S: NOTE / CHORD
B: TRACK ROOT
T: QUANT [0-100%]
S: MICRO [±48]
B: TRACK SCALE
RPT + K1–4 LIVE
K1
NR: RATE
K2
NR: SWING
T: VOLUME [0-127]
S: VELOCITY
B: TEMPO [40-200]
T: GATE [1/8-4S]
S: LEN [1/8-32S]
B: SWING [50-75%]
360° ENDLESS · ×4 ACCEL
HOLD STEP + LATER PAD = NOTE LEN [1–32S] · HOLD STEP + RPT = RATCHET · REC + STEP = CHORD
HOLD RPT = TIMING · HOLD BANK = TRACKS [SRC+DST = COPY · +STOP = MUTE] · ⇧+REC SAVE / ⇧+PLAY LOAD
S01 SW 75%
⇧ TRANS −1
PAD 13 · C3
S02 FILL
⇧ TRANS +1
PAD 14 · C#3
S03 SYNC
⇧ OCTAVE −1
PAD 15 · D3
S04 TAP
⇧ OCTAVE +1
⇧15+16 = RST
S05 SW 50%
⇧ VEL 1·SFT
G# · H.MIN
S06 SW 56%
⇧ VEL 2·MED
A · P.MAJ
S07 SW 62%
⇧ VEL 3·HRD
A# · P.MIN
S08 SW 68%
⇧ VEL 4·MAX
B · BLUES
S09 1/16
⇧ PRESET 5
E · PHRYG
S10 1/16T
⇧ PRESET 6
F · LYDIAN
S11 1/32
⇧ PRESET 7
F# · MIXOL
S12 1/32T
⇧ PRESET 8
G · LOCRIAN
S13 1/4
⇧ PST 1·PRF
C · CHROM
S14 1/4T
⇧ PST 2·DAW
C# · MAJOR
S15 1/8
⇧ PRESET 3
D · MINOR
S16 1/8T
⇧ PRESET 4
D# · DORIAN
960 PPQN ENGINE · 16 TRACKS × 64 STEPS · USB-C + BLE + 3.5MM TRS MIDI OUT · AUTO-SAVE ON STOP · FW ≥ 925
SEC-03 // TRANSPORT
MODE & SEQ CONTROLS
BT
BLE OTA
HOLD: BLE MIDI + OTA
BANK
TRKS COPY
TAP: VIEW · HLD: TRACKS
KNOB
PAT BANK
KNOB BANK (LED = PAT)
PG <
⇧ + < UNDO
STEP PAGE < · ⇧ UNDO
PG >
⇧ + > REDO
STEP PAGE > · ⇧ REDO
PLAY
⇧+PAD LOAD
PLAY / CONT · ⇧+PAD LOAD
STOP
×2 SPP 0
STOP+SAVE · ×2 REW SPP0
REC
⇧+PAD SAVE
LIVE REC · ⇧+PAD SAVE
SHFT
+PAD MENU
⇧+PAD MENU · +K3/4 SCL
RPT
⇧+RPT SEQ
HLD: TIMING · ⇧ SEQ MODE

AN/SMC-925 // SHEET 2 OF 2: 360° PINWHEEL SIDE BANDS & I/O WELL INSERT — ALIGN FRAME: 250.00 × 165.00 MM

CNR-BR
BTN SIDE
SEAM ⟶ ▶
R10.7 · 90°

### REAR-01 // HARDWARE SCHEMATIC [JIELI AC636N · BR25]

- 16× VEL PADS — MUX PB2/3 → PA4/6
- 8× ENC + 10× BTN — SPI2 595 → PB0/PC1
- 3.7V LI-ION BAT — 704060 · 2000mAh

**AC636N · BR25**
pi32v2 · 960 PPQN
PI32V2 · 512K NOR
RAM ISR SAFE

- 16× WS2812 RGB — SPI1 PB1 · 800kHz
- TRS MIDI OUT — UART1 PC5 · 31.25k
- USB-C + BLE MIDI — 3-CBL · AE40 OTA

### REAR-02 // FLASH MAP, SYSEX & SAFE-MODE RECOVERY

| | |
|---|---|
| FLASH 512K: | 0x00000 CODE · 0x30000 16×PAT · 0x50000 VM · 0x77000 PST |
| USB 3-CBL: | CBL0: Master MIDI · CBL1: Private Sinco · CBL2: Mackie |
| DIAG SYSEX: | F0 7D 53 50 01/02/05 F7 (status / counters / crash dump) |
| SAFE MODE: | Power-cycle 2× in 5s → skips pads/keys, keeps BLE + OTA! |
| UBOOT ROM: | USB_KEY handshake on DP/DM enters BR25 Mask-ROM loader |

PWR · CHG · USB-C · MIDI OUT

5V ⎓ 500mA · RED=CHG / GRN=FULL · TRS-A MIDI OUT

▲ REAR BULKHEAD (222.43×23.80mm) ▼ FRONT OPERATOR CHEAT SHEET (222.43×23.80mm)

CNR-FL
KNOB SIDE
⇧+RPT=SEQ
FW ≥ 925

### F1 // STEP & NOTE LENGTH

| | |
|---|---|
| TAP PAD: | Step On / Off (S01..S16) |
| HOLD+PAD: | Note Length 1–32 steps! |
| HOLD+KNOB: | Step P-Lock (K3 moves chord) |
| HOLD+RPT: | Ratchet ×1 → ×2 → ×3 → ×4 |
| PG < / >: | Page 1–4 · ⇧+< / > Undo/Redo |

### F2 // TRACKS [HOLD BANK]

| | |
|---|---|
| TAP BANK: | Step View ↔ Play View |
| HOLD+PAD: | Select active Track 1–16 |
| SRC+DST: | Hold Src+Tap Dst = Copy Notes |
| +⇧ COPY: | Full Copy (+Ch / Note / Vol) |
| +STOP+PAD: | Mute Trk · +⇧+STOP = Clear |

### F3 // LIVE REC & TIMING

| | |
|---|---|
| REC+PLAY: | Live Rec (hold = gate len) |
| CHORDS: | Hit active step (up to 4 notes) |
| QUANTIZE: | K4 (0–100%) non-destructive! |
| HOLD RPT: | P1-8 Rate · P9-13 Swg · P14 Fill |
| SYNC/TAP: | P15 Ext Clock · P16 Tap Tempo |

### F4 // 16 PATTERNS & CUE

| | |
|---|---|
| ⇧+REC+PAD: | Save Slot 1–16 (no pad=cur) |
| ⇧+PLAY+PAD: | Load Slot (cues next bar!) |
| SLOT LEDS: | Off=Empty · Dim=Used · Lit=Cur |
| FEEDBACK: | Blinks 3× GRN=OK / RED=ERR |
| AUTO-SAVE: | STOP saves edits to flash |

▼ LEFT ENCODER SIDE (142.43×23.80mm)

CNR-BL
REAR SEAM
SEAM ⟶ ▶
R10.7 · 90°

### LEFT-01 // K1–K8 KNOB MATRIX

| | | | |
|---|---|---|---|
| K1 | Vol 0–127 | Velocity | BPM 40–200 |
| K2 | Gate 1/8–4 | Len 1/8–32 | Swing 50–75 |
| K3 | Base Note | Note/Chord | Trk Root |
| K4 | Quantize | Micro ±48 | Trk Scale |
| K5 | Prob % | Step Prob | — |
| K6 | Ratch Rate | Stp Ratch | Trk Colour |
| K7 | Steps 1–64 | Trig Cond | MIDI Ch |
| K8 | Grid Rate | Step Tie | Sync ±120ms |

### LEFT-02 // TRIG, GATE & RATCHET

| | |
|---|---|
| TRIG 1–6: | P1:ALW · P2-3: 1:2, 2:2 · P4-6: 1..3:3 |
| TRIG 7–13: | P7-10: 1..4:4 · P11:1ST · P12:FILL · 13:!F |
| GATE [K2]: | 1/8..7/8, 1..4S · Step Len: 1..32S |
| RATCH [K6]: | AUTO(split) · 1/8 · 1/16 · 16T · 32 · 64 |
| GRID [K8]: | Keeps timing (dim=off-grid) · ⇧+K8 Speed |

▼ REAR I/O WELL INSERTS (PRIMARY + SPARE, 46.00×12.80mm)

POWER · CHG · USB-C · MIDI

POWER · CHG · USB-C · MIDI

▼ RIGHT TRANSPORT SIDE (142.43×23.80mm)

CNR-FR
FRONT SEAM
STOP×2=0
R10.7 · 90°

### RIGHT-01 // ROOTS & SCALES [P1–12]

| | |
|---|---|
| P01: C · Chromatic | P07: F# · Mixolydian |
| P02: C# · Major | P08: G · Locrian |
| P03: D · Nat Minor | P09: G# · Harm Minor |
| P04: D# · Dorian | P10: A · Pent Major |
| P05: E · Phrygian | P11: A# · Pent Minor |
| P06: F · Lydian | P12: B · Blues |

0 LIT = FOLLOW · BANK K3/4 = TRK · ⇧+K3/4 = GLOBAL

### RIGHT-02 // LED READOUTS & CLOCK

| | |
|---|---|
| BAR LEDS: | Vol / Vel / Prob / Qnt / BPM: P1→16 |
| SIGNED: | Micro & Sync: P8..1 (-) · P9..16 (+) |
| NOTE KEY: | P1–12 = C..B Root · P13–16 = Octave |
| EXT SYNC: | P15 On = 24 PPQN · STOP×2 = SPP 0 |
| SYNC MS: | BANK K8: −120..+120ms (Master/Slave) |

SCALE CHECK: 100.00 MM VERNIER

0 10 20 30 40 50 60 70 80 90 100

360° PINWHEEL WRAP:
1. Top Deck (flush inside rim)
2. Rear → Left (butt at SEAM)
3. Front → Right (butt at SEAM)
4. I/O Insert (inside rear well)

1:1 SCALE · US LETTER 8.5×11

CAMEO 5: Cut RED (#ff0000) paths · Set BLUE (#0066ff) 250×165mm Align Frame to NO CUT after centering.